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Comparative Study of Using Different Electric Motors in the Electric Vehicles

Nasser Hashemnia and Behzad Asaei
Dept. of Electrical and Computer Eng., University of Tehran
E-mail: naserhashemn@ut.ac.ir

***Abstract*—In this paper, different electric motors are studied and compared to see the benefits of each motor and the one that is more suitable to be used in the electric vehicle (EV) applications. There are five main electric motor types, DC, induction, permanent magnet synchronous, switched reluctance and brushless DC motors are studied. It is concluded that although the induction motors technology is more mature than others, for the EV applications the brushless DC and permanent magnet motors are more suitable than others. The use of these motors will result in less pollution, less fuel consumption and higher power to volume ratio. The reducing prices of the permanent magnet materials and the trend of increasing efficiency in the permanent magnet and brushless DC motors make them more and more attractive for the EV applications.**

I. INTRODUCTION

Selection of the traction motor for an electric vehicle system is a crucial step in designing the overall system. Many criteria such as efficiency, cost, reliability, power density, maturity of technology and controllability must be taken into consideration. In the industrial application point of view, the most common motors used in the hybrid electric vehicles (HEV) and pure electric vehicles (PEV) are: DC motors, induction, permanent magnet synchronous, switched reluctance and brushless DC motors [1]. The cross sections of these motors are shown in Fig. 1.

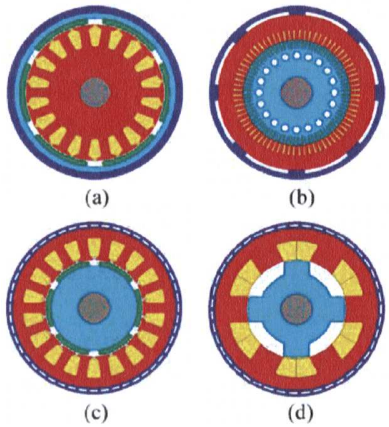

Fig. 1. Different electric motors for use in traction motors: (a)- DC motor, (b)- Induction motor, (c)- Permanent magnet brushless motor and (d)- Switched reluctanc motor.

AC motor drives have some distinct advantages over their DC counterparts, such as: higher efficiency, higher power density, effective regenerative braking, robustness, reliability and less need of maintenance [2]. DC motors are losing their attraction while switched reluctance motors are "the dark horse in the race" [3]. In [4], a survey and comparison of characteristics of motor drives used in the EVs are presented. Regarding the proportion of research, the other three types including the induction, brushless DC and permanent magnet synchronous motors are greatly dominant. Therefore, in this paper these motors are used in EV applications and the vehicles with same characteristics but various motors are simulated by Advisor® software in different driving cycles to see and compare them in terms of pollution, fuel consumption, and power to volume ratio.

II. COMPARATIVE STUDY

In this section, the advantages and disadvantages of different electric motors are discussed.

A. DC motors

Although DC motors have been the subject of interest since old time because of simple control and decoupling of flux and torque, their construction (having brushes and rings) poses maintenance problems. Therefore, after the growth of vector control for AC motors (synchronous and induction), the DC motors' attraction in traction applications diminished.

Of course, DC motors are still good candidates for low power applications. The commutator actually acts as a robust inverter; Therefore, power electronics devices can be mush simple and inexpensive. The Peugeot factory of France has introduced a HEV named "Dynavolt" in which, DC motor has been used as traction motor.

B. Induction motors (IM)

Squirrel cage induction motors have already been the most important candidate because of their reliability, robustness, less maintenance and the ability to work in hostile environments. The induction motors have the most mature technology among all other AC competitors. In Fig. 2, the main characteristics of an induction motor have been shown. Torque and field control can be decoupled using vector control methods. Speed range may be extended using flux weakening in the constant power region.

Existence of break-down torque in the constant power region, reduction of efficiency and increment of losses at high speeds, intrinsically lower efficiency in comparison to

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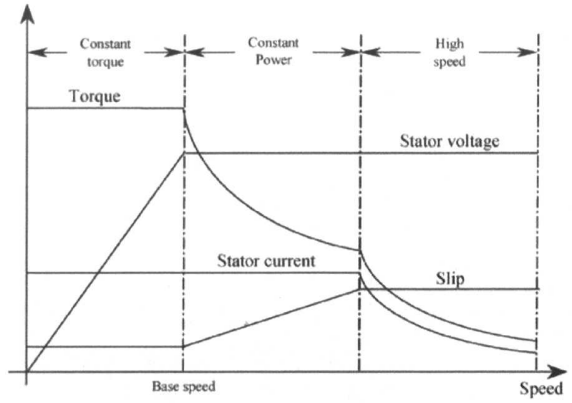

Fig. 2. Different characteristics of induction motors

permanent magnet motors due to the presence of rotor winding and finally low power factor are among the shortcomings of induction motors. Many efforts have been made by researchers to solve these problems, such as: usage of dual inverters to extend the constant power region, incorporating doubly- fed induction motors to have excellent performance at low speeds and reducing rotor winding losses at the design stage.

C. *Permanent magnet synchronous (PMS) motors (or brushless AC (BLAC))*

PMS motors are the most serious competitor to the induction motors in traction applications. Actually, many car manufacturers (such as Toyota, Honda and Nissan) have already used these motors in their vehicles. These motors have several advantages: higher power density, higher efficiency and the more effective distribution of heat into the environment. However, these motors have intrinsically a narrow constant power region (Fig. 3-a). To widen the speed range and increase the efficiency of PMS motors, conduction angle of the power converter can be controlled at speeds higher than the base speed. Fig. 3-b shows the torque speed of a PMS motor with conduction angle control. Speed range can be extended to three of four times the base speed.

A shortcoming of these motors is that they can be demagnetized due to the heat or armature reaction.

D. *Switched reluctance motors (SRM)*

Switched reluctance motors are receiving much attraction in HEV systems every day. Among the advantages of these motors are: simple and rigid construction, fault tolerance, simple control and excellent torque-speed characteristic. A switched reluctance motor can intrinsically operate under a wide constant power region. Several disadvantages such as high noise, high torque ripple, special convertor topology and electromagnetic interference have been mentioned for this motor [1]. Both the advantages and disadvantages of this motor are important in the EV applications. A conventional torque-speed characteristic of a SRM has been depicted in Fig. 4.

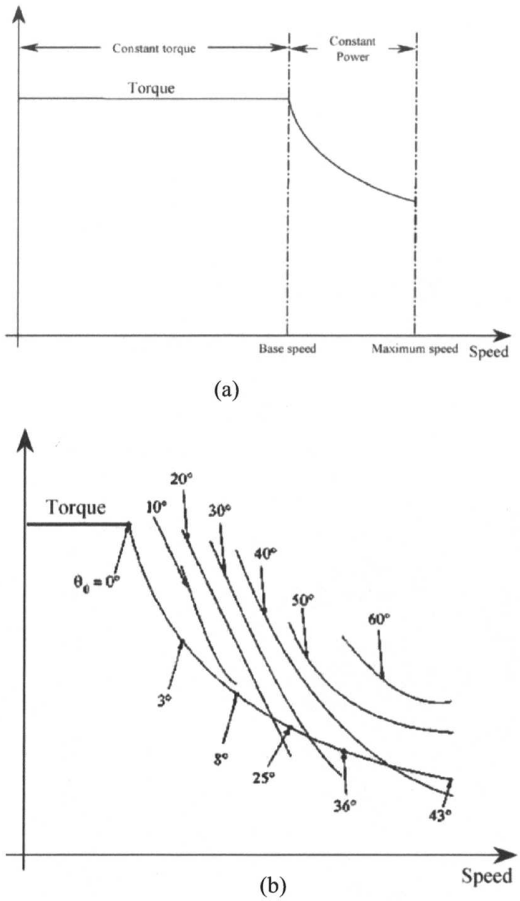

Fig. 3. Torque- speed characteristic of BLAC motor: (a)- conventional characteristic and (b)- characteristic under conduction angle control.

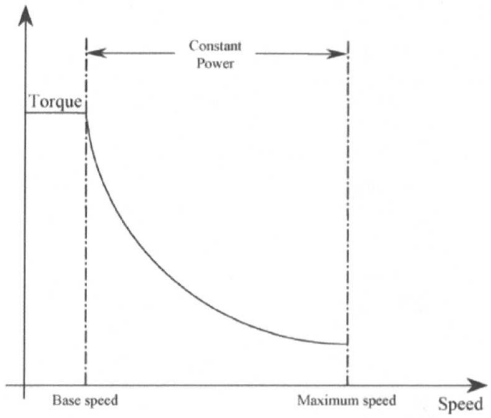

Fig. 4. Conventional characteristic of a SRM

E. Brushless DC motors (BLDC)

These motors are conceptually the outcome of reversing the stator and rotor of permanent magnet DC motors. They are fed by rectangular waves in contrast to BLAC motors which are fed by sinusoidal waves. Their main advantages are the deletion of the brushes, their compactness, high efficiency and high energy density [4].

In table I, common EVs and their propulsion systems have been shown.

In [1], the traction systems commonly used in EVs are evaluated based on six factors. As shown in table. II, a score out of 5 is given for each point to each motor. It is concluded that based on these factors, the IM and PM motors are more suitable. However, in the following section the DC and SRM motors are not taken into consideration due to their disadvantages.

TABLE I

COMMON EVS AND THEIR PROPULSION SYSTEMS

HEV Model	Propulsion System
PSA Peugeot-Citroën/Berlingo (France)	Dc Motor
Holden/ECOmmodore (Australia)	Switched Reluctance Motor
Nissan/Tino (Japan)	Permanent Magnet Synchronous Motor
Honda/Insight (Japan)	Permanent Magnet Synchronous Motor
Toyota/Prius (Japan)	Permanent Magnet Synchronous Motor
Renault/Kangoo (France)	Induction Motor
Chevrolet/Silverado (USA)	Induction Motor
DaimlerChrysler/Durango (Germany/USA)	Induction Motor
BMW/X5 (Germany)	Induction Motor

TABLE II [1]

EVALUATION OF DIFFERENT TRACTION SYSTEMS FOR ELECTRIC VEHICLES

Propulsion Systems / *Characteristics*	DC	IM	PM	SRM
Power Density	2.5	3.5	5	3.5
Efficiency	2.5	3.5	5	3.5
Controllability	5	5	4	3
Reliability	3	5	4	5
Technological maturity	5	5	4	4
Cost	4	5	3	4
Σ Total	**22**	**27**	**25**	**23**

As can be observed, PM motors have the highest power density due to the presence of high power density permanent magnets. Moreover, they have the highest efficiency because of the absence of rotor losses. DC and induction motors have the best controllability and their flux and torque control can be easily decoupled. The induction motor has the best reliability due to its robust and rigid construction.

Using reference [2], different characteristics of electric motors have been compared in table. III.

TABLE III

COMPARISON OF DIFFERENT CHARACTERISTICS OF ELECTRIC MOTORS

Efficiency, %

	Motor only	Electronics	Motor &&Electronics
PM	97	93	90
SRM	94	90	85
IM	90	93	84
DC	80	98	78

Weight (Based on PM=100)

	Motor only	Motor && Electronics
PM	100	200
SRM	150	250
IM	200	300
DC	400	450

Comparative cost (including electronics) (DC=100)

	1993	1998	2003
DC	100	105	110
IM	100	90	80
SRM	150	90	70
PM	150	90	60

III. Simulation Results

In this part, three major electric vehicles, namely IM, PM and BLDC, are studied by simulation software Advisor®. Simulation is performed under three different driving cycles, namely: CYC-UDDS, Constant Speed, and NurembergR36 to compare the fuel consumption and the air pollution of the vehicle if these motors are used. The characteristics of these cycles are listed in Table. IV. Efficiency map for different operating points is needed for each motor. This characteristic has been shown in Fig. 5 using reference [3]. Tables V through VII show the simulation results.

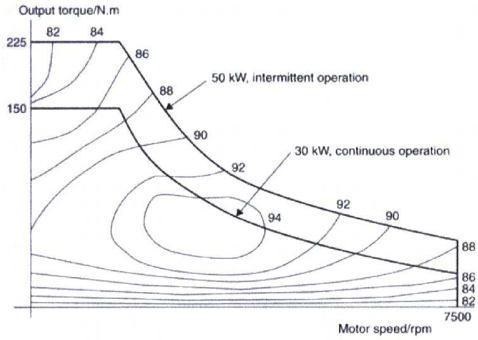

Fig. 5. Efficiency map for a BLDC motor [3]

TABLE IV
CHARACTERISTICS OF THE THREE CYCLES

Cycle	CYC-UDDS	Constant 60	NurembergR36
V_{max}(km/h)	91.25	96.56	53.7
V_{avg}(km/h)	31.51	96.56	14.33
$A_{accel,avg}$(m/s^2)	0.5	0	0.58
$A_{dec,avg}$(m/s^2)	-0.58	0	-0.55

TABLE V
CYC-UDDS CYCLE

Motor	IM	PM	BLDC
HC(gr/km)	0.168	0.16	0.158
Co(gr/km)	0.819	0.738	0.729
Nox(gr/km)	0.185	0.168	0.167
Consumption(L/100km)	6.8	6.4	6.3

TABLE VI
CONSTANT SPEED CYCLE

Motor	IM	PM	BLDC
HC(gr/km)	0.168	0.16	0.158
Co(gr/km)	0.819	0.738	0.729
Nox(gr/km)	0.185	0.168	0.167
Consumption(L/100km)	6.8	6.4	6.3

TABLE VII
NOREMBERGR36 CYCLE

Motor	IM	PM	BLDC
HC(gr/km)	0.168	0.16	0.158
Co(gr/km)	0.819	0.738	0.729
Nox(gr/km)	0.185	0.168	0.167
Consumption(L/100km)	6.8	6.4	6.3

The output of the software for the BLDC motor case has been shown in Fig. 6. The operating points for the engine of the vehicle in the torque- speed space have been depicted in Figs. 7- 9. As can be seen, the brushless DC and the permanent magnet motors have a rather good priority over induction motors in terms of fuel consumption and pollution. As brushless DC and permanent magnet motors are not so much different with respect to fuel consumption and pollution, a detailed comparison with respect to motor control, cost, starting torque and performance at high speeds is necessary.

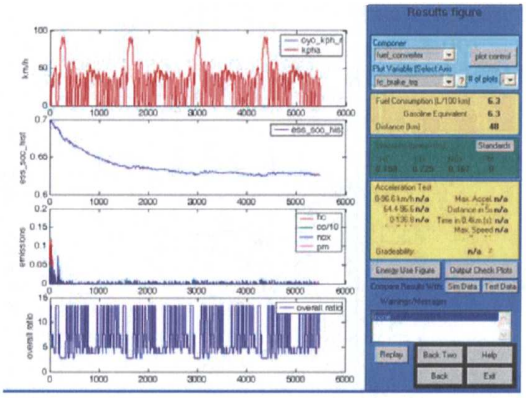

Fig. 6. Output of the software for the BLDC motor

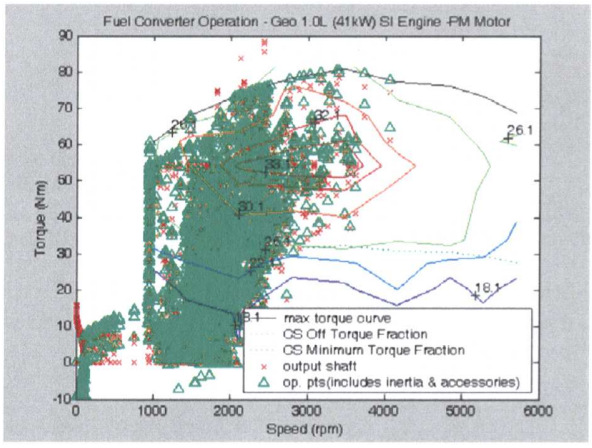

Fig. 7. Operating points of the engine for the PM motor

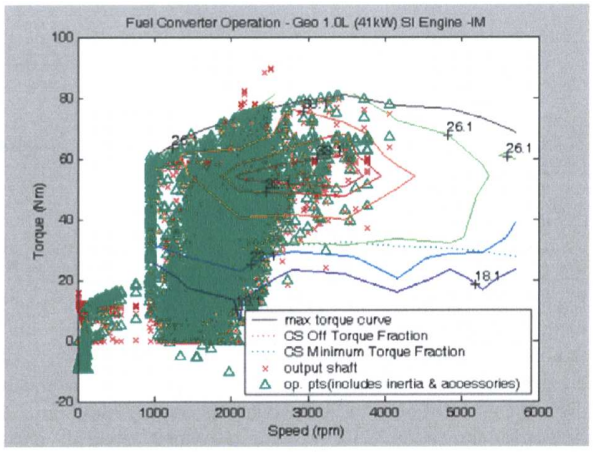

Fig. 8. Operating points of the engine for the induction motor

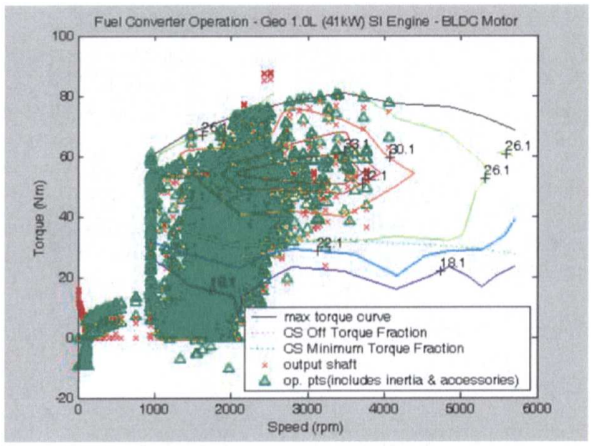

Fig. 9. Operating points of the engine for the BLDC motor

IV. CONCLUSION

The induction motors have been known as the best candidate for the EV applications because they are robust, less costly, mature in technology and need less maintenance. However, in this paper it is demonstrated that in terms of pollution and fuel consumption, the permanent magnet and the brushless DC motors have more priorities such as less pollution, less fuel consumption and more power to volume ratio which makes them attractive in the EV applications.

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